Three Tier Architecture on AWS

Terraform deployment with public, application, and data tiers across two Availability Zones.

By: Mos

Presentation Plan

Architecture

Diagram, tier purpose, traffic flow, design choices.

Live Demo

ALB URL, instance rotation, health endpoint, isolation test.

Costs

Monthly estimate, cost controls, savings opportunities.

Challenges

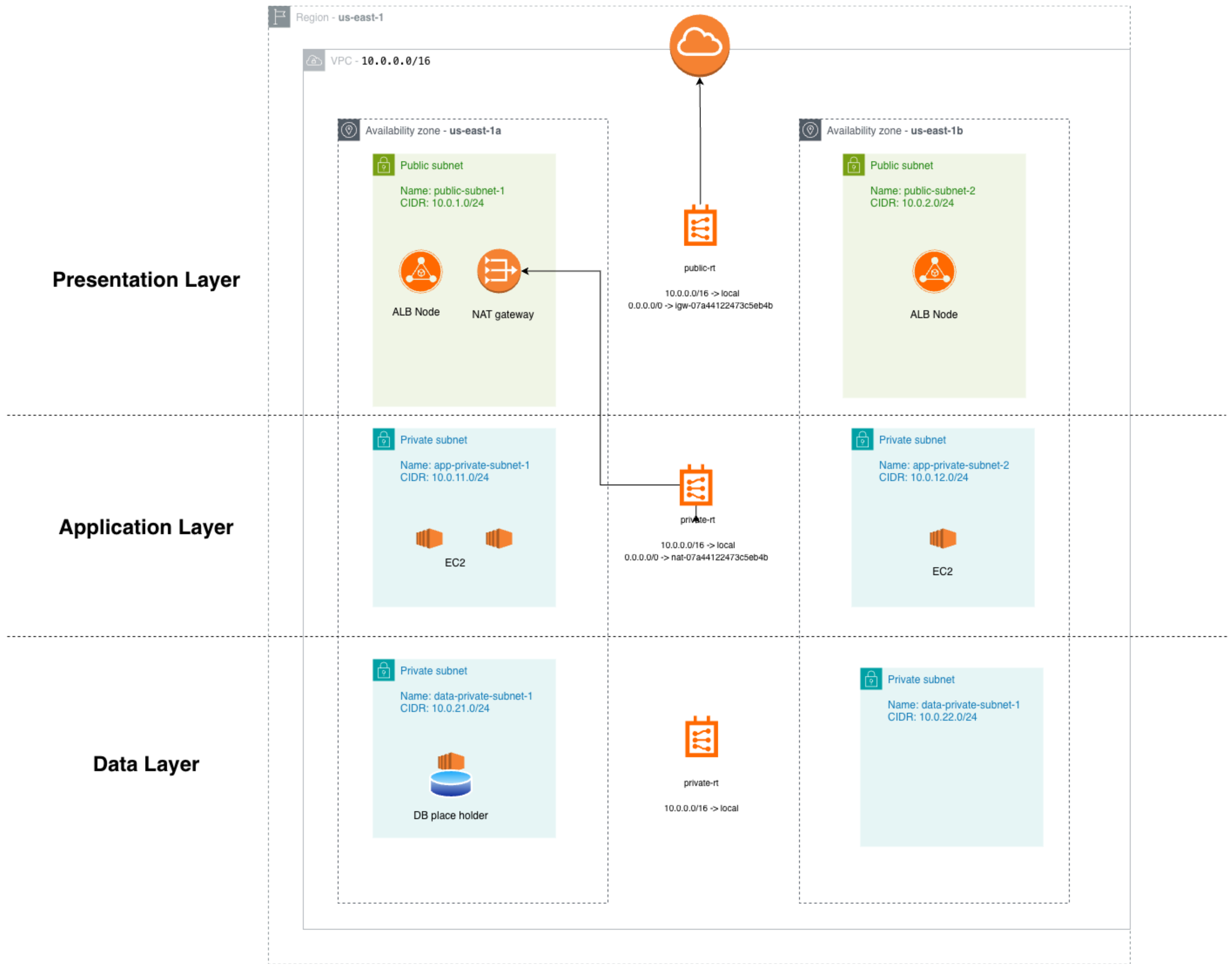
Issues encountered and how they were solved.

Next Steps

Production improvements, scale, limitations.

Q&A

Architecture Overview



Tier Responsibilities

Presentation Tier

- Internet facing ALB
- HTTP listener on port 80
- Health checks on `/health`
- Public subnets in two AZs

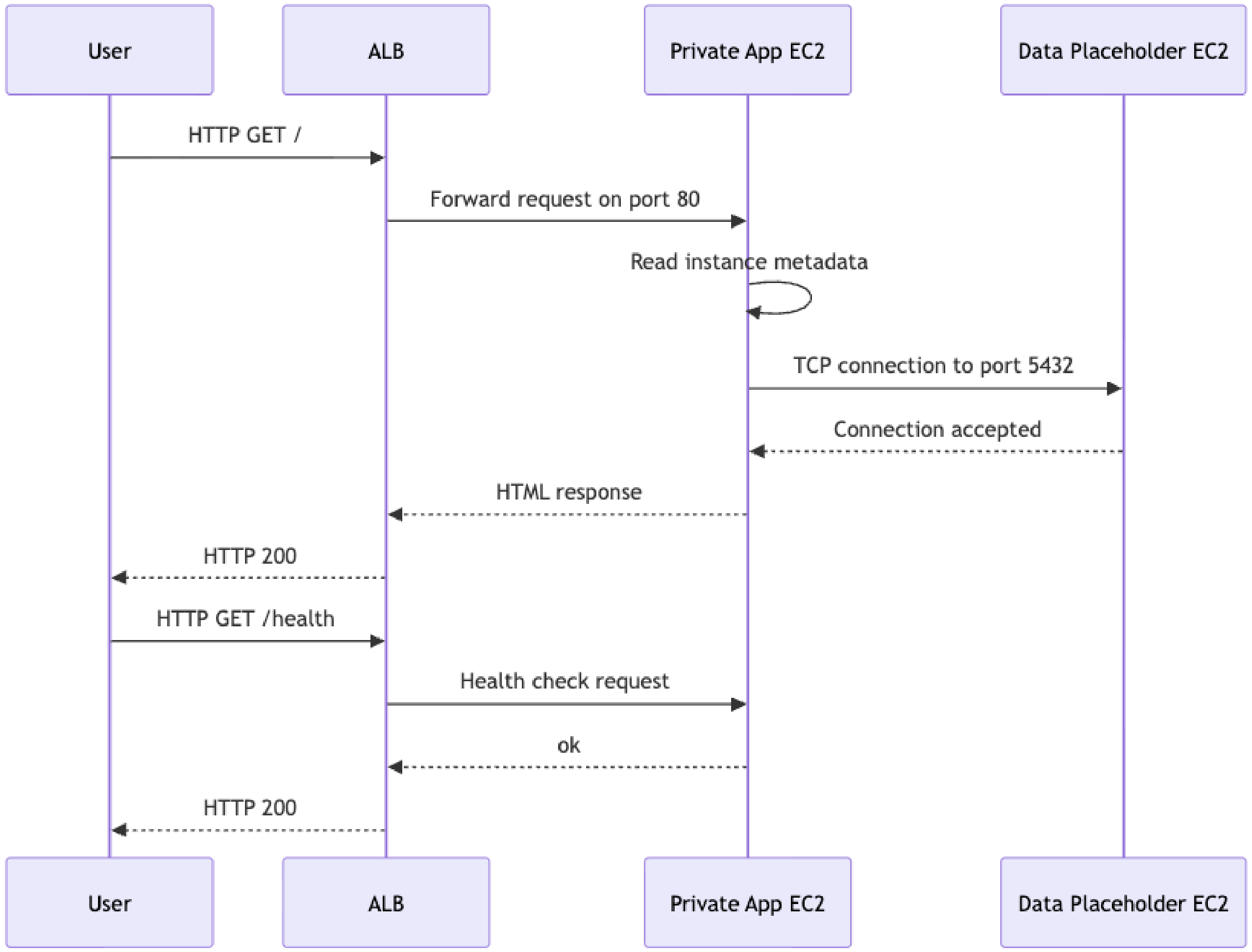
Application Tier

- Three private EC2 instances
- Returns instance ID, private IP, AZ, and DB status
- Registered in ALB target group
- Outbound internet through NAT Gateway

Data Tier

- Private EC2 database placeholder
- Listens on port 5432
- Only app security group can connect
- No default internet route

Traffic Flow



Request Path

- User sends HTTP request to the ALB DNS name.
- ALB forwards traffic to a healthy app instance.
- App reads EC2 metadata for instance ID, private IP, and AZ.
- App checks connectivity to the database placeholder.
- Health checks use `/health` and return `ok`.

Key Design Decisions

Network

- VPC CIDR: 10.0.0.0/16
- Six subnets across two AZs
- Separate route tables for public, app, and data tiers
- Data tier route table has local VPC route only

Security

- ALB is the only public entry point
- App allows traffic only from ALB security group
- Data allows port 5432 only from app security group
- IMDSv2 required and root volumes encrypted

Live Demo

Steps:

- Open the app through the ALB DNS name.
- Refresh or run curl to show different instance IDs.
- Open `/health` and show `ok`.
- Try SSH to a private IP to show isolation.

```
ALB_DNS=app-alb-1196878497.us-east-1.elb.amazonaws.com

curl http://$ALB_DNS/health

for i in {1..20}; do
  curl -s http://$ALB_DNS | awk -F'[<>]'
'/class="value">i-/{print $3}'
done | sort | uniq -c
```

Demo Results

Load Balancing

```
7 i-0041fb351f4d31891
6 i-02f121d4d79e00783
7 i-03768b6cd9c713ae6
```

20 requests were distributed across all three app instances.

Failover

```
Stopped:
i-0041fb351f4d31891

After stop:
10 i-02f121d4d79e00783
10 i-03768b6cd9c713ae6
```

The ALB removed the stopped instance and kept serving traffic.

Challenges and Solutions

Challenges

- Keeping the cost low with the NAT Gateway and ALB requirement.
- Stopping and starting instances will kill the running process.

Solutions

- Used Terraform to be provision and destroy the infrastructure as needed.
- Ensure linux services are configured to start on boot.

Cost Analysis

Item	Estimate	Note
4 t3.micro EC2 instances	$30.37/month	Before Free Tier
32 GB gp3 EBS	$2.56/month	Before Free Tier
1 NAT Gateway	$32.85/month	Largest fixed cost
1 Application Load Balancer	$16.43/month	Before LCU usage
Fixed subtotal	$82.21/month	Before traffic based charges
Planning range	$80-$100/month	Low traffic demo estimate

$82.21

Fixed monthly subtotal from EC2, EBS, NAT Gateway hourly, and ALB hourly costs.

$80-$100

Planning range with a small buffer for usage based charges.

For this demo, the main cost is not traffic. The main fixed cost is the NAT Gateway hourly charge.

Cost Optimization

Already Applied

- Used t3.micro instances.
- Used small 8 GB gp3 root volumes.
- Used EC2 database placeholder instead of RDS for the lab.
- Used one NAT Gateway instead of one per AZ.

Savings Opportunities

- Destroy stack after demo or testing.
- Use one app instance while developing.
- Remove NAT Gateway when it is not needed.
- Use schedules or Auto Scaling for non demo hours.

Improvements and Next Steps

Production Additions

- HTTPS with ACM certificate.
- AWS WAF on the ALB.
- CloudWatch logs, metrics, and alarms.
- Remote Terraform state with locking.
- Secrets Manager for database credentials.

Scale and HA

- Use Auto Scaling Group for app instances.
- Replace placeholder DB with RDS Multi AZ.
- Add one NAT Gateway per AZ for production HA.
- Add backup and restore.
- Use Route 53 and a custom domain.

Q&A